

Construction & Calibration of a low cost Polarimeter

1. Background

Measurement of concentration of glucose in a solution is crucial in pharmaceutical, food and the beverage industry. Polarimetry is a well established optical method to find the concentration of optically active substances such as glucose by measuring the rotation of plane-polarized light.

Biot's law of optical rotation states that the observed angle of rotation (α) is directly proportional to the concentration (c), the path length (l), and the specific rotation ($[\alpha]$) of the substance in solution:

$$[\alpha] = \alpha / c \cdot l$$

Malus's law describes how the intensity of light transmitted through a polarizing filter changes depending on the angle between the filter's axis and the light's polarization direction. The polarization direction of an electromagnetic radiation is considered as the direction of oscillation of the electric field. Where I is the intensity of light after passing through the polarizing filter while I_0 is the intensity before passing through the filter.

$$I = I_0 \cos^2(\theta)$$

1.1 Research Problem

Determining glucose concentration accurately is crucial for industries with strict quality control requirements. Commercial polarimeters provide high precision but are expensive and difficult to maintain, making them unsuitable for small-scale industries. Affordable alternatives like DIY polarimeters often lack precision, consistency, and reliability.

The main challenge in developing a DIY polarimeter is achieving industrial-level accuracy while keeping costs low. Factors such as light source stability, precise angle measurement, and calibration methods need improvement for better performance.

This study aims to tackle these challenges by building and refining a DIY polarimeter based on Biot's law to measure glucose concentration accurately. The research will focus on enhancing

light polarization, path length control, and digital detection to create a cost-effective and precise solution for industrial use.

1.2 Literature

Optical rotation is the rotation of plane-polarized light by chiral molecules, depending on the substance's concentration, path length, and light wavelength. While traditional polarimeters provide high precision, they can be costly and inaccessible. Recent advancements favor cost-effective setups using LEDs for wavelength-specific light and digital tools like smartphone cameras for recording. Studies highlight the feasibility of low-cost, portable devices using polarizers, solution containers, and rotatable analyzers for educational and research purposes. This experiment adopts these principles, integrating LEDs, polarizers, and a smartphone camera in a simple, robust design to study optical rotation effectively and affordably.

2. Research Objectives:

1. Design and construct a cost-effective, portable polarimeter using a circular LED array, optical polarizers, and a smartphone camera.
2. Investigating the relationship between the angle of optical rotation and the concentration of glucose solution at different wavelengths.
3. Evaluate the accuracy and reliability of the low-cost polarimeter by comparing experimental data with theoretical predictions and established benchmarks.
4. Demonstrate the feasibility of using a simple, affordable setup for educational purposes and basic research in optical rotation.

3. Methods:

3.1 Device:

The device will feature a circular LED array with five different wavelengths, a solution container (16 cm × 3 cm × 5 cm), two polarizers, and a phone camera for recording results. The LED array will be placed as a light source where it will be collimated using a slit made in cardboard and convex lens. The sugar solution will be placed between the polarizers, with the second polarizer mounted on a rotatable holder to measure the angle of rotation. All components will be housed in

a sturdy apparatus to ensure alignment and minimize external interference. The device will illuminate the solution with polarized light, and the phone camera will record changes in polarization at different wavelengths.

3.2 Sample

A set of glucose solutions is prepared within the concentration range of 1.25 to 3.5 mol dm⁻³, with 5 different solutions, each having a volume of 150 mL.

3.3 Analysis

The analysis involves finding the angle of twisting. By using Malu's law and a smartphone camera to capture the intensity of light, that video is analyzed frame by frame using python and OpenCV to find the point of minimum intensity. Then the obtained concentration vs the twisting angle can be used to create a transfer function. Then the concentration of the unknown glucose solution can be found.

4. Ethical & Safety concerns

Soldering LED arrays involves safety risks such as inhaling toxic solder fumes and burns from the hot soldering iron. To ensure safety, work in a well-ventilated area, wear protective equipment such as goggles, use lead free solder to prevent the formation of toxic fumes, and handle tools with care to prevent injuries and fire hazards. The usage of a helping-hand to hold the components to be soldered can minimize the risk of injuries.

5. Budget & duration

Total cost: LKR 5000

Duration: 5 weeks

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